Financial NLP and Sentiment Analysis

Lecture 09 — Treating Language as Financial Data

Juan F. Imbet

Paris Dauphine – PSL University

June 20, 2026

Outline

Why Treat Language as Data?

THE BIGGER PICTURE

A single trading day now generates hundreds of thousands of news articles, social posts, filings, and analyst reports. Extracting the signal that moves prices — in real time, repeatably — is a defining problem of quantitative finance this decade.

Text is a data source with its own **measurement challenges**, exactly as:

- price data has bid–ask bounce,
- volume data has reporting lags,
- **text** has selection, duplication, and timestamp pitfalls.

This lecture

Where financial text comes from → how to score its sentiment → what each source predicts → how to turn scores into a tradeable signal → the production pipeline.

Five Sources of Financial Text

Sources differ on four axes: **frequency**, **curation**, **audience**, **regulatory oversight**.

Source	**Curation / latency**	**Canonical finding**
Social media	none / real-time	Bollen 2011: Twitter mood $\rightarrow$ DJIA
News & wires	high / ms–hours	Tetlock 2008: neg. words $\rightarrow$ returns
Earnings calls	scripted + Q&A	Larcker 2012: tone $\rightarrow$ restatements
SEC filings	legal / quarterly	Loughran–McDonald 2011
Central banks	extreme / scheduled	Schmeling 2019: tone $\rightarrow$ equities

Source shapes method

The structural properties of each source determine both the information it carries *and* the right preprocessing / model. There is no one-size-fits-all pipeline.

Source Properties That Matter

Social media (X, Reddit/WSB, StockTwits)

- Least curated; high noise, sarcasm, bots
- Captures *retail attention* orthogonal to institutions
- Aggregates inform even when posts don't

News & wires

- Reuters/Bloomberg: ms latency, structured event
- Key flaw: **selection** — only newsworthy firms

Earnings calls

- Prepared remarks = soft-info framing
- Q&A reveals analyst concerns + candour

SEC filings (EDGAR): 10-K (Item 1A, MD&A), 10-Q, 8-K, DEF 14A

- Disclosure law ⇒ truthful but vague

Central banks: FOMC, ECB

- Meaning in calibrated *deviations* from boilerplate

Preprocessing: From Raw Text to Tokens

UNDER THE HOOD

Every pipeline normalises, deduplicates, and time-aligns before modelling. Skipping any of these biases the downstream sentiment measure.

Tokenization & normalization

- Subword tokenizers: BPE (GPT) / WordPiece (BERT).
- Case folding for lexicons; *preserve* case for cased transformers.
- Number handling (`[NUM]`) — but EPS \$2.40 vs \$0.24 carries signal.
- Cashtag/ticker harmonisation (\$AAPL); boilerplate removal (XBRL, exhibits).

Preprocessing: Deduplication & Temporal Alignment

Deduplication — most important, most neglected

- Wires reprint verbatim; retweets propagate identical text.
- Duplicates bias sentiment *and* inflate sample size $\rightarrow$ spurious narrow CIs.
- MinHash / LSH; mark dupes when trigram Jaccard > 0.6–0.8.

Temporal alignment — avoid look-ahead bias (post-close news $\nrightarrow$ same-day return) and stale timestamps; store publication *and* retrieval times separately.

Three Coexisting Generations

THE BIGGER PICTURE

No single method dominates. Lexicons are fast and interpretable; fine-tuned classifiers are strong in-domain; LLMs are flexible and need no labels. A competent practitioner uses all three and knows when each fits.

Method	**Labels?**	**Strength**	**Weakness**
Lexicon (LM)	none	transparent, cheap	no negation/contex
Fine-tuned (FinBERT)	yes	strong in-domain	needs labelled data
LLM zero/few-shot	none	flexible, nuanced	cost & latency

Lexicon Method: Loughran–McDonald

UNDER THE HOOD

Net sentiment ratio for document d:

$$S_{\text{LM}}(d) = \frac{N_{\text{pos}}(d) - N_{\text{neg}}(d)}{N_{\text{words}}(d)}$$

Why a finance dictionary? Harvard General Inquirer mislabels *tax, liability, cost, foreign, crude, cancer* as negative — neutral in 10-Ks. On 10-Ks, GI negativity tracks legal boilerplate, not performance.

LM word lists (built from the 10-K vocabulary):

Negative	Litigious	Positive	Uncertainty	Weak modal	Strong modal
2,355	903	354	297	27	19

The **negative** list dominates — the asymmetry of disclosure language.

LM Worked Example & Limits

MD&A excerpt

"Revenue declined 12%, reflecting challenging conditions and competitive pressure. The Company remains cautious but is confident in its long-term position."

- Negative hits: *declined, challenging, pressure, cautious* $= 4$
- Positive hits: *confident* $= 1$; total non-trivial words ≈ 35
- $S_{\text{LM}} = (1-4)/35 \approx -\mathbf{0.086}$ (bottom-quartile MD&A)
- A GI classifier also flags *competitive* $\rightarrow \approx -0.17$: a distortion.

Known limits

No negation ("not confident" scores positive); context-blind; domain shift (misses "headwinds", "softness"). Still widely used: transparent, cheap, auditable.

Fine-Tuned Classifiers: FinBERT & SEC-BERT

UNDER THE HOOD

Take a pre-trained encoder, attach a head, fine-tune. For the [CLS] embedding $\mathbf{h} \in \mathbb{R}^{d_h}$:

$$P(y = c \mid d) = \mathrm{softmax}(\mathbf{W}_c\mathbf{h} + \mathbf{b}_c), \quad c \in \{\text{pos}, \text{neu}, \text{neg}\}$$

minimising cross-entropy; predict $\hat{y} = \arg\max_c P(y = c \mid d)$.

FinBERT (Araci 2019; Huang 2023)

- Further-pretrain BERT on financial news + filings
- Fine-tune on PhraseBank + analyst + call data
- Beats GI, LM, and generic BERT

SEC-BERT (Loukas 2022)

- Pre-trained *only* on EDGAR (10-K/Q/8-K)
- Vocabulary tuned to legal-financial text
- Wins in-domain on risk-factor / event tasks

Edge over lexicons: $\mathbf{h}$ sees the whole sequence $\Rightarrow$ handles negation

The Financial PhraseBank

The standard labelled benchmark (Malo et al. 2014):

- **4,840** news sentences, labelled pos / neu / neg.
- **16** finance-trained annotators; majority label used.
- At 75% agreement: $\approx$ 3,400 high-confidence sentences, inter-annotator agreement $\approx$ 75–80%.
- Neutral $\approx$ half the corpus — motivates macro-F1, not accuracy.

Caveat

News-sentence style $\neq$ filings / call language. Models tuned on PhraseBank may transfer poorly to long, syntactically complex documents — supplement with domain-matched labels.

LLM Zero-Shot & Few-Shot Classification

THE BIGGER PICTURE

GPT-4 / Claude / LLaMA-3 classify sentiment from a task description alone — *no fine-tuning, no labelled set*. Rapid prototyping on new domains.

Zero-shot prompt sketch: *"You are a financial analyst. Classify the sentiment … from the perspective of a stock investor."* → "Neutral" with a rationale (revenue growth offset by margin compression).

LLM Classification: Benchmarks & Cost

Benchmarks (real findings):

- Ko & Lee 2024: GPT-4 zero-shot *matches or exceeds* FinBERT on PhraseBank, FiQA, and a proprietary call dataset — with no labels.
- López-Lira & Tang 2023: GPT headline scores positively predict next-day returns, beating lexicons.
- Kirtac & Germano 2024: OPT (GPT-3 family) yields a markedly higher long–short Sharpe than the LM dictionary.
- Fatemi & Hu 2023: fine-tuned *small* LLMs hit SOTA; extra few-shot examples beyond a small set add no accuracy.

Cost is the catch: $\sim$\$100–\$500 per million short sentences $\Rightarrow$ distil into a small model for production.

One Source, One Research Question

THE BIGGER PICTURE

Each source has a natural question: does retail mood drive short-run prices? does news carry information beyond prices? does management tone reveal private information? does disclosure language predict performance? do policy words move asset prices?

Source	**Question**
Social media	Retail attention $\rightarrow$ price pressure?
News	Incremental information beyond prices?
Earnings calls	Tone $\rightarrow$ private information?
SEC filings	Disclosure language $\rightarrow$ future risk/return?
Central banks	Linguistic cues $\rightarrow$ asset prices?

Social Media: Attention, Disagreement, Bank Runs

Evidence:

- Bollen 2011: "calm" Twitter mood Granger-causes DJIA (10-week sample; mixed replications).
- Cookson & Niessner 2020: StockTwits *disagreement* predicts volume and spreads.
- Gil-Bazo et al. 2025: 1.6M fund-family posts — tone *and* volume predict fund inflows.
- Cookson et al. 2026: SVB collapse — high Twitter-exposure banks lost more value; social media can *catalyse* bank runs (streaming use case).

WSB / GameStop (Jan 2021): coordinated WSB sentiment led GME/AMC by 1–2 days; mechanism via call demand $\rightarrow$ gamma-hedging amplification.

Methodological hazards

Bots & coordinated inauthentic behaviour; severe selection; reverse causality (returns $\rightarrow$ posting).

News, Earnings Calls, SEC, Central Banks

News (Tetlock 2007/08)

- WSJ pessimism $\rightarrow$ price pressure + reversion
- Firm-level neg. language $\rightarrow$ earnings
- Stale negativity adds no forecast power

Earnings calls

- Larcker 2012: deception cues $\rightarrow$ restatements
- Price 2012: tone $\rightarrow$ post-call abnormal returns
- Cook & Kazinnik 2023: local LLMs $\approx$ closed-source, lower privacy risk

SEC filings

- LM negativity $\downarrow$ filing-day return, $\uparrow$ volume
- Cohen 2020 "lazy prices": unchanged 10-Ks underperform
- Lehner 2024: LLMs can rewrite filings to shift measured tone

Central banks

- Schmeling 2019: positive tone $\rightarrow$ equities $\uparrow$
- Hansen 2018: deliberation quality $\rightarrow$ inflation outcomes
- Nakamura 2018: information vs. policy effect confound

MD&A Readability: The Fog Index

UNDER THE HOOD

$$\text{Fog} = 0.4\left(\frac{N_{\text{words}}}{N_{\text{sentences}}} + 100 \cdot \frac{N_{\text{complex}}}{N_{\text{words}}}\right)$$

N_{complex} = words with ≥ 3 syllables (excl. proper nouns, compound modifiers, simple verb inflections).

- Fog $= 12 \Leftrightarrow$ US high-school level.
- 10-K MD&A typically **18–22** — much harder than a newspaper.
- Li 2008: higher Fog $\rightarrow$ lower current earnings, higher earnings volatility (managers obscure when performance is poor).
- Lehavy 2011: less readable 10-Ks $\rightarrow$ fewer, less accurate analysts.

Raw Scores Are Not Yet a Signal

THE BIGGER PICTURE

A score per document is not tradeable. It needs aggregation, denoising, and statistical transformation. Two empirical frameworks test economic content: event studies and predictive regressions.

Document $\rightarrow$ entity aggregation (all docs on firm i, day t):

- Simple mean: $\bar{S}_t = \frac{1}{n_t} \sum_i S_i$
- Volume-weighted (reach / circulation) · Recency-weighted (exp. decay)

Noisy sources (social) need weighting; curated sources (filings) do not.

Standardization & Stationarity

UNDER THE HOOD

Cross-sectional z-score within industry-year removes sector language fixed effects:

$$\tilde{S}_{i,t} = \frac{S_{i,t} - \mu_{j(i),t}}{\sigma_{j(i),t}}$$

($j(i)$ = industry of firm i.) A defence firm and a software firm become comparable.

Stationarity

- Long sentiment series drift (style change, universe change, regime shift).
- Apply ADF / KPSS; first-difference or detrend non-stationary series.

Sentiment inflation

Re-scoring old text with a newer model changes measured sentiment. Use one *fixed* model vintage across all periods — or document the vintages.

Event Studies & Predictive Regression

UNDER THE HOOD

Abnormal return $AR_{i,\tau} = R_{i,\tau} - \hat{R}_{i,\tau}$ (benchmark, e.g. CAPM). Cumulative over $[a, b]$:

$$\mathrm{CAR}_i[a,b] = \sum_{\tau=a}^{b} AR_{i,\tau}, \qquad \overline{\mathrm{CAR}}[a,b] = \tfrac{1}{N}\sum_i \mathrm{CAR}_i[a,b].$$

Does text beat the EPS number? Condition on the earnings surprise:

$$\mathrm{CAR}_i[1,60] = \alpha + \beta_1\,\mathrm{SUE}_{i,0} + \beta_2\,\tilde{S}_{i,0} + \varepsilon_{i,0}$$

with $\mathrm{SUE} = (\mathrm{EPS} - \widehat{\mathrm{EPS}})/\hat{\sigma}$. Incremental text content $\Rightarrow \hat{\beta}_2 > 0$.

High-frequency FOMC: 30-min window around announcements (Nakamura 2018) demands sentence-level scoring *inside* the window $\rightarrow$ low-latency inference.

Two Horizons, Two Mechanisms

	Short horizon	**Long horizon**
Channel	attention / price pressure	fundamental information
Evidence	Da 2015: attention $\rightarrow$ reversal	Tetlock 2008; Cohen 2020
Data needs	real-time, low-latency	daily/weekly, heavier models
Audience	HFT-adjacent industry	most academic research
Alpha	larger (pre-cost)	smaller, more accessible

Practical implication

Short-horizon strategies need real-time feeds and execution within minutes; long-horizon strategies tolerate batch processing and richer models. Industry chases the short horizon (more alpha); academia studies the long horizon (lower data/execution cost).

From “What” to “How”

THE BIGGER PICTURE

Earlier sections said what to extract and why. The pipeline is the engineering: collect text at scale, run inference in production, evaluate classifier quality with metrics that don’t mislead.

Three production concerns:

1. **Data collection** — EDGAR, social APIs, news vendors.
2. **Inference at scale** — batch (throughput) vs. streaming (latency).
3. **Evaluation** — label quality *and* classifier quality *and* economic validity.

(Collection & scale engineering detailed in the appendix.)

Evaluation: Annotator Agreement

UNDER THE HOOD

Krippendorff's alpha measures agreement beyond chance:

$$\alpha = 1 - \frac{D_o}{D_e}$$

D_o = observed, D_e = expected disagreement. Nominal k-class: $D_e = 1 - \sum_c (n_c/n)^2$. $\alpha = 0$: chance; $\alpha = 1$: perfect.

Thresholds: $\alpha \geq 0.667$ reliable; ≥ 0.800 high confidence. Financial benchmarks: **0.65–0.78** — genuine ambiguity of finance sentences.

Dataset	Classes	Alpha
Financial PhraseBank (100% agr.)	3	1.00
Financial PhraseBank (66% agr.)	3	≈ 0.72
Earnings call sentences (practitioner)	3	0.68–0.75
FOMC statement sentences	3	0.70–0.78

Evaluation: Classifier Metrics & Economic Validity

UNDER THE HOOD

Per class c:

$$\mathrm{P}_c = \frac{TP_c}{TP_c + FP_c}, \ \mathrm{R}_c = \frac{TP_c}{TP_c + FN_c}, \ F1_c = \frac{2\,\mathrm{P}_c\mathrm{R}_c}{\mathrm{P}_c + \mathrm{R}_c}.$$

Report **macro-F1** (unweighted mean over classes).

Why macro, not accuracy

Neutral $\approx$ half of financial sentences. An accuracy-maximiser that over-predicts neutral scores high while failing on the economically important positive / negative classes.

Economic validity $\neq$ statistical metrics. Macro-F1 $= 0.75$ can still have zero alpha if the test set is unrepresentative or labels are misaligned. Best practice: regress aggregate sentiment on forward returns / earnings

Lecture 09 Summary

- **Sources:** social $\rightarrow$ filings, spanning curation and oversight; structure dictates preprocessing and model.
- **Methods:** lexicon (LM, the filings standard), fine-tuned (FinBERT/SEC-BERT), LLM zero/few-shot — LLMs match or beat fine-tuned without labels, at higher cost.
- **Applications:** attention (social), information (news), soft disclosure (calls), risk (filings), policy signal (central banks).
- **Signal:** aggregate $\rightarrow$ z-score within industry-year $\rightarrow$ test stationarity; event study + predictive regression assess economic content.
- **Pipeline:** batch + streaming inference; evaluate with Krippendorff's α, macro-F1, *and* a held-out economic test.

Next

Practical 09 — LM scoring, FinBERT vs. lexicon, and a filing-date event study on real EDGAR + price data.

Appendix

Data Collection, Inference at Scale, and Extensions

Extra / optional material — beyond the core lecture.

Appendix: Data Collection

EDGAR full-text search (`efts.sec.gov`)

1. Query form types (10-K, 8-K) and date range.
2. Download index pages → document links.
3. Parse primary HTML/SGML (`BeautifulSoup`, `sec-edgar-downloader`).
4. Extract MD&A / Risk Factors by header matching.
5. Store Parquet, indexed by CIK / period / filing date.

Twitter/X Academic API (paid since 2022): filters `has:cashtags`, `lang:en`, verified-author. Alternatives: StockTwits API, Reddit Pushshift (restricted), commercial sentiment feeds.
News vendors: Bloomberg, Dow Jones DNA, Refinitiv, RavenPack (machine-readable event records); GDELT (free, noisier, 15-min updates) for research.

Appendix: Inference at Scale

Batch (throughput-bound):

- FinBERT (110M), 512 tokens, single A100: hundreds of docs/sec; multi-million archive in hours.
- Optimisations: dynamic padding/batching; FP16/BF16; distillation (110M $\rightarrow$ 33M, $< 5\%$ accuracy loss); data/CPU parallelism.

Streaming (latency-bound):

- Target < 100 ms for FOMC; near-real-time for live call transcription.
- Kafka/Kinesis $\rightarrow$ ONNX inference workers $\rightarrow$ downstream aggregation.
- `onnxruntime` + quantised weights: < 100 ms/sentence on modern CPUs.

Appendix: Ordinal Krippendorff's Alpha

For ordered labels (pos $>$ neu $>$ neg) the nominal coincidence metric is replaced by a weighted disagreement:

$$d^2(c, c') = \Big(\sum_{g=c}^{c'} n_g\Big)^2,$$

penalising disagreements by the number of scale steps between labels (n_g $=$ frequency of category g). Ordinal α is more conservative than nominal α for sentiment corpora — a stricter lower bound on agreement.

Appendix: NVIX, Factor Models, ESG News

News-Implied Volatility (Manela & Moreira 2017): project WSJ front-page text onto realized VIX variation; the measure extends to **1890**, capturing disaster-risk eras (Depression, WWII, 2008–09) far beyond VIX's 1990 start.

News in factor models (Ke et al. 2019): supervised extraction of price-relevant news features adds predictive content, esp. at short horizons; intraday alpha captured by HFT, longer-horizon signals lower-Sharpe but accessible.

Climate / ESG (Engle et al. 2020): a climate-news index builds a hedge portfolio; high climate-news exposure $\rightarrow$ lower returns on high-news days. Generalises to pandemic / geopolitical / supply-chain risk indices.

Appendix: Domain-Adapted LLMs on Long Filings

Beyond the context window:

- Chiu & Hung 2024: finance-specific LLaMA-2 + LLM summarisation of long 10-Ks → higher buy-and-hold abnormal returns than FinBERT.
- Siano 2025: LLM word-context embeddings of earnings press releases explain far more return variation than dictionaries; raise R^2 of financial-statement surprises.
- Xu & Babaian 2025: FinBERT + GPT-4o *ensemble* on FOMC press conferences — cross-family combination is more stable than any single model.